

EEE 416 (January 2025)

Microprocessors and Embedded Systems Laboratory

Final Project Report

Section: B2 Group: 04

Design and Implementation of an ESP32 Based Wireless Sensor Network for Real-Time Logistics and Cold Chain Monitoring

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1. Abstract

This project presents the design and implementation of an intelligent, IoT-enabled vaccine carrier for real-time cold chain monitoring. The system addresses the critical need for maintaining strict temperature controls for vaccines during transportation to prevent spoilage and ensure efficacy. The core of the solution is a dual ESP32-based system integrated with a suite of sensors (BME280, MPU-6050), a GPS module (NEO-8M) for location tracking and a GSM module (SIM800L) for communication. The primary function of the system is the continuous, real-time monitoring of environmental parameters including temperature, humidity and atmospheric pressure, as well as physical parameters like orientation and location. In case of any breach in the predefined safe conditions (e.g., temperature exceeding the 2-8°C range), the system instantly sends an alert via SMS to relevant stakeholders and updates a web dashboard. This portable, cost-effective solution aims to significantly reduce vaccine wastage, enhance logistics management, and improve the overall efficiency of healthcare delivery, particularly in regions with limited infrastructure.

2. Introduction

The integrity of the vaccine cold chain is paramount to global public health. Vaccines are highly sensitive biological products that require strict temperature control, typically between 2°C and 8°C, to maintain their potency and efficacy. A failure in the cold chain can render vaccines ineffective and unsafe. The World Health Organization (WHO) estimates that up to 25% of vaccines arrive at their destination damaged due to exposure to improper temperatures.

Current monitoring practices often rely on manual checks and passive data loggers, which lead to significant delays in detecting temperature breaches. This lack of real-time information means that corrective action cannot be taken promptly, resulting in substantial vaccine wastage. Furthermore, the absence of real-time tracking complicates logistics and accountability.

This project's scope is to design and develop a portable, cost-effective and intelligent system for monitoring vaccine carriers during transportation. By integrating environmental sensors, GPS and GSM technology, the system provides a comprehensive, real-time solution. The expected impact is a marked reduction in vaccine wastage, improved efficiency in healthcare delivery and enhanced public health outcomes by ensuring that potent vaccines reach communities, including those in remote areas.

3. Design

3.1 Problem Formulation

3.1.1 Identification of Scope

- **Primary Scope:** Real-time monitoring of vaccine storage conditions (temperature, humidity, pressure) during transport.
- **Physical Monitoring:** Orientation detection using an accelerometer/gyroscope to assess handling and prevent damage from drops.
- **Location Tracking:** GPS-based real-time position monitoring for logistics management.
- **Communication:** An immediate alert system via SMS and a web dashboard to notify healthcare workers of any breaches.
- **Power Management:** A portable, battery-operated solution designed for extended use.

3.1.2 Literature Review

The project is grounded in existing research on wireless sensor networks (WSN), cold chain logistics, and IoT applications. Key areas reviewed include:

- **Cold Chain Studies:** Research on the impact of temperature breaches on vaccine efficacy highlights the critical need for precise monitoring.
- **ESP32 Applications:** The use of ESP32 microcontrollers in low-power WSN and IoT projects is well-documented, demonstrating their suitability for this application.
- **Sensor Fusion & GSM Integration:** Techniques for combining data from multiple environmental sensors and integrating GSM modules for remote monitoring and alerts were analyzed to inform the system's design.

The review identified a research gap in affordable, portable and fully integrated real-time monitoring solutions specifically tailored for vaccine logistics.

3.1.3 Formulation of Problem

The core problem is the need for a reliable, real-time, and portable system to monitor vaccine conditions during transit. This breaks down into several key challenges:

1. **Continuous Monitoring:** Ensuring consistent and accurate environmental monitoring in dynamic, mobile conditions.
2. **Immediate Notification:** Designing a system for instant breach detection and notification.
3. **Logistics Management:** Integrating location tracking for efficient logistics and asset management.

4. **Power Efficiency:** Developing a power-efficient system for extended operational periods.
5. **Cost-Effectiveness:** Creating a solution that is affordable for widespread deployment.

3.1.4 Analysis

- **System Requirements:** The system must provide real-time monitoring, wireless communication, and be housed in a portable, robust design.
- **Performance Metrics:** Key metrics include sensor accuracy, alert response time, battery life, and communication reliability.
- **Constraints:** The design is constrained by power consumption, physical size, and overall cost.
- **Feasibility:** A feasibility study confirmed the availability of suitable components, the manageable complexity of development, and the practical viability of deployment.

3.2 Design Method

The system employs a dual-controller architecture to distribute tasks and optimize performance.

- **Master Device (ESP32-S3):** This is the main controller responsible for acquiring and processing data from all connected sensors (BME280, MPU-6050, NEO-8M). It runs the core logic for threshold monitoring.
- **Slave Device (TTGO T-Call ESP32):** This secondary controller, featuring an integrated SIM800L module, is dedicated to handling GSM communication. It receives breach alerts from the master device and is responsible for sending SMS notifications.
- **Inter-Controller Communication:** The two ESP32 devices communicate using the ESP-NOW protocol, a low-power, direct device-to-device communication method that does not require a Wi-Fi router. The master sends a signal to the slave upon detecting a breach.

The operational logic is based on four key principles:

1. **Threshold-Based Monitoring:** The ESP32-S3 continuously compares sensor readings against preset limits (e.g., temperature $< 2^{\circ}\text{C}$ or $> 8^{\circ}\text{C}$).
2. **Sensor Fusion:** Data from multiple sensors (temperature, humidity, pressure, motion) are combined to provide a comprehensive view of the package's condition.
3. **GPIO Signaling / ESP-NOW:** Upon detecting a breach, the master controller signals the slave controller to trigger the SMS alert.
4. **Sleep Mode Optimization:** Power management algorithms are implemented to put the controllers into a low-power sleep mode during periods of inactivity, extending battery life.

3.3 Block Diagram

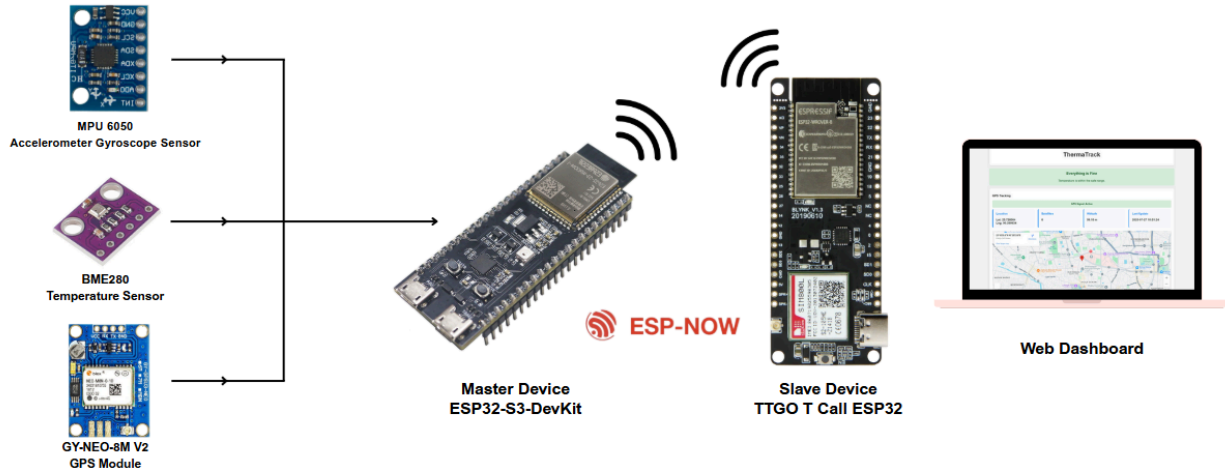


Fig-1: A block diagram illustrating the system architecture.

The ESP32-S3 (Master) is connected to the BME280, MPU-6050, and NEO-8M GPS module via I2C and UART interfaces. It is powered by the 10,000 mAh power bank. The TTGO T-Call ESP32 (Slave) is also powered by the power bank and communicates with the master via ESP-NOW.

3.3 Hardware Design



Fig-2 : An image of the physical prototype inside the thermal insulated box.

The hardware is housed within a standard thermal insulated vaccine carrier box. The electronic

components—including the microcontrollers, sensors, and power bank—are mounted on a breadboard and securely affixed to the underside of the box's lid. This placement ensures the sensors accurately measure the internal environment while protecting the electronics. The design is self-contained and portable.

4. Implementation

4.1 Description

The implementation involved assembling the hardware, developing the firmware for both microcontrollers, and integrating the system.

- **Hardware Assembly:** The ESP32-S3, TTGO T-Call, BME280, MPU-6050, and NEO-8M were connected on a breadboard according to the circuit design. Antennas for GSM and GPS were connected to their respective modules. The entire assembly was powered by a 10,000 mAh power bank and fitted inside the insulated box.
- **Firmware Development:**
 - **Master (ESP32-S3):** The firmware was written in C/C++ using the Arduino IDE. It initializes all sensors, reads data at regular intervals, checks for threshold breaches, and sends alerts to the slave device via ESP-NOW.
 - **Slave (TTGO T-Call):** The firmware for the slave device initializes the SIM800L module, listens for incoming ESP-NOW messages from the master, and upon receiving an alert, constructs and sends an SMS message containing breach details and GPS coordinates to a predefined phone number.
- **Web Dashboard:** A web dashboard was developed to visualize the data in real-time, showing current sensor readings, GPS location on a map, and system status.

5. Design Analysis and Evaluation

5.1 Novelty

The novelty of this project lies in:

- **Integrated Solution:** It combines environmental, motion, and location monitoring into a single, cohesive system.
- **Dual-Controller Architecture:** It uses specialized controllers for sensing and communication to improve reliability and efficiency.
- **Real-Time Breach Detection:** The system provides immediate alerts, enabling timely intervention.
- **Cost-Effective Approach:** It utilizes affordable, off-the-shelf components, making it a

viable solution for widespread deployment.

- **Wireless Communication:** It uses GSM for alerts, ensuring functionality even in areas without internet access, and ESP-NOW for efficient inter-device communication.

5.2 Design Considerations

- **Accuracy:** Sensors were calibrated and validated to ensure measurement accuracy within their specified operating ranges.
- **User Experience:** The system is designed for simple operation with clear status indicators on the web dashboard.
- **Maintainability:** A modular design allows for easy replacement of individual components.
- **Environmental Robustness:** The system is designed to operate reliably in the varying temperature and humidity conditions encountered during transport.

5.3 Investigations

A series of investigations and tests were conducted to validate the system's performance.

- **Sensor Characterization:** The accuracy and drift of the BME280 and MPU-6050 sensors were analyzed.
- **Communication Testing:** GSM signal strength and SMS delivery reliability were tested in various locations. The range and reliability of the ESP-NOW communication were also validated.
- **Power Consumption Analysis:** The system's power draw was measured under different operating conditions to estimate battery life. The 10,000 mAh power bank was estimated to last for approximately 7 days of continuous operation.
- **GPS Accuracy:** The precision of the NEO-8M GPS module was tested in both open and partially obstructed environments.
- **Integration Testing:** The full system was tested to ensure proper inter-module communication and synchronization. An 8-hour endurance test showed the system was stable, with breach alerts sent in approximately 5 seconds. A 6-hour real-world simulation was completed with 100% success.

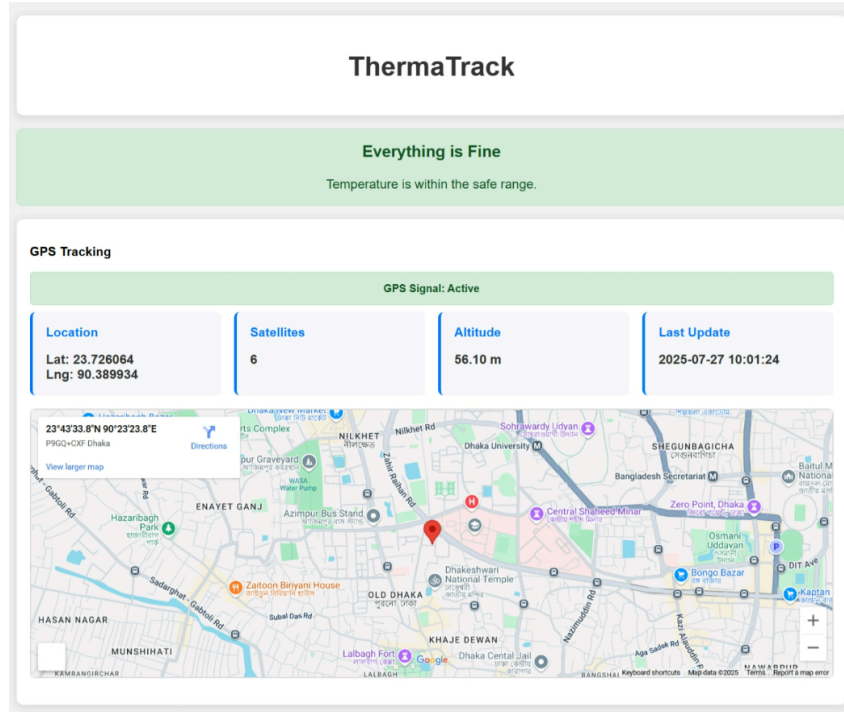


Fig-3: Web dashboard showing the map and sensor data.



Fig-4: Graphs showing Temperature, Humidity, and Pressure data from the web dashboard.

5.4 Limitations of Tools

The system's performance is subject to the limitations of its components:

Component / Factor	Limitation	Operating Temperature (°C)
BME280 Sensor	Precision is limited at extreme temperatures.	-40 to 85
GPS Signal	Accuracy is reduced in indoor or obstructed environments.	-40 to 85
GSM Coverage	Performance is dependent on cellular network availability.	-40 to 85
Power Bank	Limited battery life requires periodic recharging.	-20 to 60
ESP32 (Processing)	Has computational limitations for complex algorithms.	-40 to 85
Memory (16MB Flash)	Limited data storage capacity.	-40 to 85
SMS Communication	Delivery times can be variable, causing delays.	-40 to 85

5.5 Impact Assessment & Sustainability Evaluation

- **Healthcare & Social Benefits:** The project directly contributes to public health by reducing vaccine wastage and improving immunization program effectiveness, especially in remote areas.
- **Economic Impact:** The system offers significant cost savings by preventing vaccine spoilage.
- **Regulatory & Scalability:** It provides verifiable data for regulatory compliance and is adaptable to other cold chain applications (e.g., food, pharmaceuticals).
- **Global Health:** The solution supports global health initiatives by strengthening vaccine distribution networks in developing regions.

6. Reflection on Individual and Team work

6.1 Individual Contribution of Each Member

Team Member	Primary Responsibilities
Kaidul	Implemented ESP-NOW communication, ESP32-S3 firmware development, sensor data acquisition, hardware design, system integration, environmental sensor calibration, and performance validation.
Wasif	Physical enclosure design, thermal insulation setup, component selection and procurement, TTGO T-Call programming for SMS alerts, and power distribution circuit design.
Sadaf	ESP32-S3 firmware development, threshold monitoring algorithm implementation, system testing, GPS module integration and coordinate parsing, and TTGO T-Call programming for SMS alerts.
Shiam	TTGO T-Call GSM module programming, SIM800L AT command implementation, management of wireless communication protocols, real-time alert system development, and ESP-NOW communication implementation.

6.2 Mode of Team Work

The team adopted a highly collaborative and agile approach throughout the project lifecycle. We established a clear workflow from the outset, which included weekly team meetings to discuss progress, brainstorm solutions to challenges and assign tasks for the upcoming week. Task allocation was based on the individual strengths and interests of each member, ensuring high-quality work and engagement. For day-to-day communication, a dedicated Telegram group was used for quick queries and updates, while a shared Google Drive served as a central repository for all project documents, datasheets, and code. For more complex tasks, such as integrating the firmware for the two microcontrollers, we employed pair programming sessions to enhance code quality and facilitate knowledge sharing. This structured yet flexible methodology fostered a productive environment, ensuring the project remained on schedule and all members contributed effectively.

6.3 Diversity Statement of Team

The success of this project was significantly amplified by the diverse skill sets and perspectives each member brought to the team. Our group comprised a balanced mix of expertise in hardware, software, and systems integration. Kaidul and Shiam provided deep knowledge in embedded systems programming and wireless communication protocols, which was essential for the core functionality. Wasif's proficiency in hardware design, component sourcing, and power systems was critical in constructing a reliable and robust physical prototype. Sadaf's strengths in algorithmic logic and sensor data processing were key to implementing the accurate monitoring and alert systems. This blend of technical abilities allowed us to tackle multifaceted challenges efficiently. Every member's unique viewpoint was valued in our discussions, leading to more innovative problem-solving and a final product that is more comprehensive and well-engineered than what any individual could have achieved alone.

6.4 Log Book of Project Implementation

Date	Main Activities	Outcomes & Results
14.05.2025	Finalized and presented the project proposal.	Project approved for implementation.
22.05.2025	Created detailed system architecture and selected components.	Completed system architecture and component list.

29.05.2025 - 19.06.2025	Procured components and performed initial hardware testing.	Components received and basic functionality verified.
26.06.2025	Integrated the sensor system into the thermal insulated box.	Sensor system successfully set up inside the box.
03.07.2025	Developed firmware (Phase 1) for ESP32-S3.	Core logic for sensor reading and breach detection completed.
10.07.2025	Developed firmware (Phase 2) for TTGO T-Call.	Successful SMS alerts. Estimated battery life of 7 days.
18.07.2025	Performed full system integration and comprehensive testing.	Breach alert sent in ~5s. System stable during 8-hour test.
24.07.2025	Conducted advanced mobile and real-world simulation tests.	System proven stable. 100% success in 6-hour simulation.
29.07.2025	Completed final report, presentation, and demonstration.	Project completed on time. All requirements met.

7. Communication to External Stakeholders (PO(j))

7.1 Executive Summary

This project introduces a robust, low-cost IoT solution for real-time monitoring of vaccine carriers. Addressing the critical issue of vaccine spoilage due to temperature fluctuations in the cold chain, our system provides continuous tracking of temperature, humidity, physical orientation, and GPS location. If conditions deviate from the safe range (2-8°C), the system instantly alerts managers via SMS and a live web dashboard. This enables immediate corrective action, drastically reducing vaccine wastage and enhancing supply chain integrity. The prototype,

built with ESP32 microcontrollers and standard sensors, is a portable, battery-powered unit designed for easy integration into existing logistics. It offers a scalable and affordable solution to improve public health outcomes by ensuring vaccines are delivered safely and effectively.

7.2 User Manual

1. Initial Setup:

- Ensure the 10,000 mAh power bank is fully charged.
- Confirm that a valid, activated SIM card with credit is inserted into the TTGO T-Call module.

2. Powering On the System:

- Connect the USB cables from both the ESP32-S3 and the TTGO T-Call modules to the power bank.
- The system will automatically power on and begin initialization. This process takes approximately 30-60 seconds. During this time, the modules will connect to the cellular network and acquire a GPS signal.

3. Normal Operation:

- Once initialized, the system enters its normal monitoring state. It continuously logs data and updates the web dashboard.
- Place the sealed vaccine carrier in the transport vehicle. The device will operate autonomously.

4. Monitoring the Shipment:

- Access the web dashboard using the provided URL.
- The dashboard displays real-time data for temperature, humidity, and pressure.
- A map shows the carrier's current location, which updates periodically.

5. Alert Conditions:

- If the internal temperature goes outside the 2°C to 8°C range, or if a significant physical shock is detected, the system will automatically send an SMS alert to the pre-configured

phone number.

- The SMS will include the type of breach and the last known GPS coordinates.

6. Powering Off the System:

- At the end of the journey, simply disconnect the USB cables from the power bank.

7.3 Github Link

The complete source code for the firmware, web dashboard, and hardware schematics are available at the following Github repository:

<https://github.com/Chrollo8/ESP32-Based-Wireless-Sensor-Network-for-Real-Time-Logistics-and-Cold-Chain-Monitoring>

7.4 Video Demonstration Link

A video demonstration of the project, showcasing the hardware setup, real-time dashboard, and SMS alert functionality, can be viewed on YouTube:

https://drive.google.com/drive/folders/1M_msSI9Zwd7a1hYS4FyEzVGud9_3y0o0?usp=sharing

8. Project Management and Cost Analysis

8.1 Bill of Materials

The following table provides the bill of materials for this project.

Category	Unit	Cost (BDT)
ESP32-S3	1	1500
TTGO T-Call ESP32 Module	1	2550
BME280 Sensor	2	960
MPU-6050 (GY-521)	2	470

GPS Module (GY-NEO-8M V2)	1	1250
Thermal Insulated Box	1	900
Power Bank (10,000 mAh)	1	1300
Breadboard	1	120
Antenna for GSM	1	245
Antenna for GPS	1	345
Total		9,640

8.2 Per Unit Cost of Prototype

The following table provides the cost breakdown for the prototype unit.

Category	Unit	Cost (BDT)
ESP32-S3	1	1500
TTGO T-Call ESP32 Module	1	2550
BME280 Sensor	1	480
MPU-6050 (GY-521)	1	235
GPS Module (GY-NEO-8M V2)	1	1250
Thermal Insulated Box	1	900

Power Bank (10,000 mAh)	1	1300
Breadboard	1	120
Antenna for GSM	1	245
Antenna for GPS	1	345
Total		8,925

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

The cost of a mass-produced unit is expected to be significantly lower than the prototype due to economies of scale, bulk purchasing of components and optimized manufacturing processes. The following table provides an estimated cost breakdown for a production run of 1,000+ units.

Key Assumptions for Cost Reduction:

- **Bulk Component Purchasing:** Component costs are estimated to decrease by 30-40% when sourced in large quantities from manufacturers or distributors.
- **Custom PCB Manufacturing:** The breadboard (120 BDT) and manual wiring are replaced by a custom-designed Printed Circuit Board (PCB). While a single PCB is expensive, the per-unit cost for mass production is drastically lower.
- **Integrated Components:** For a large production run, a single custom module integrating the microcontroller, sensors, and communication hardware could be designed, further reducing costs. The estimate below assumes the use of discrete components on a custom PCB.

Estimated Mass-Production Cost (per unit):

Category	Estimated Cost (BDT)	Notes
ESP32-S3 & TTGO T-Call Modules	2,500	Combined cost with 35% bulk discount.
BME280 Sensor	300	38% bulk discount.
MPU-6050 (GY-521)	150	36% bulk discount.

GPS Module (GY-NEO-8M V2)	800	36% bulk discount.
Thermal Insulated Box	700	Bulk order discount.
Power Bank (10,000 mAh)	850	Bulk order discount.
Antennas (GSM & GPS)	350	Combined cost with 40% bulk discount.
Custom PCB & Assembly	400	Replaces breadboard, includes manufacturing cost.
Estimated Total	6,050	

8.4 Timeline of Project Implementation

Activities	Week Number (Jan 2025 Start)
Proposal Presentation	Week 5
System design, Component Selection & Procurement	Weeks 6-7
Hardware Setup, Initial Testing, Integration	Weeks 8-9
Firmware Development	Weeks 10-11
Integration and System Testing	Weeks 12-13
Final Report, Presentation, Demonstration	Week 14

9. Future Work (PO(I))

While the current prototype successfully meets the project objectives, several avenues exist for future enhancement and development:

- **Miniaturization and PCB Design:** The current breadboard-based prototype can be replaced with a custom-designed Printed Circuit Board (PCB). This would significantly reduce the overall size of the electronics, improve reliability and durability, and create a more professional, production-ready device.
- **Advanced Power Management:** Further optimization of the sleep modes and power cycles could extend the battery life beyond the current 7-day estimate. Exploring the integration of a small solar panel on the carrier's lid could enable self-charging capabilities for very long-duration shipments.
- **Cloud Platform Integration and Data Analytics:** Migrating from a simple web dashboard to a robust cloud IoT platform (such as AWS IoT, Google Cloud Platform, or Microsoft Azure) would enable advanced data analytics. This would allow for the storage of historical data, trend analysis, prediction of potential failures, and generation of comprehensive reports for entire fleets of carriers.

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